Day 2 Problem Set Solutions

Methods Camp | University of Texas at Austin

Casey Breen

Day 2 Problem Set

This document contains solutions to all exercises from the Day 2 problem set. To reinforce your understanding, please try your best to complete these exercises independently before referring to the solutions.

Exercise 1: Random Variables & Expectation

A survey asks respondents how many times they attended religious services last month. Suppose the distribution is:

Times attended (x)	0	1	2	4
$P(X = x)$	0.4	0.3	0.2	0.1

1.1 Is X discrete or continuous? Why?

Solution: X is **discrete**: it only takes on a countable set of values (0, 1, 2, 4), not any value in a continuous interval.

1.2 Compute $E[X]$ by hand.

Solution:

$$E[X] = 0(0.4) + 1(0.3) + 2(0.2) + 4(0.1) = 0 + 0.3 + 0.4 + 0.4 = 1.1$$

1.3 In your own words, what does $E[X]$ tell us?

Solution: $E[X] = 1.1$ is the long-run average across many respondents — if we surveyed thousands of people and averaged their attendance, we'd land close to 1.1.

1.4 Bonus: if attendance is recoded as $Y = 2X + 1$, what is $E[Y]$? (Use the linearity property.)

Solution: Using linearity, $E[aX + b] = aE[X] + b$:

$$E[Y] = E[2X + 1] = 2E[X] + 1 = 2(1.1) + 1 = 3.2$$

Exercise 2: Distributions

2.1 For each scenario below, name the most appropriate distribution (Bernoulli, Binomial, Poisson, or Normal) and briefly justify your choice:

a. Whether a single respondent owns a home (yes/no)

Solution: Bernoulli — a single yes/no trial for one respondent.

b. The number of home-owners among 200 randomly sampled respondents

Solution: Binomial — the count of "successes" (home-owners) across $n = 200$ independent Bernoulli trials, each with the same probability p.

c. The number of hate-crime incidents reported in a city per month

Solution: Poisson — a count of relatively rare events occurring over a fixed interval (a month), without a natural fixed number of "trials."

d. Adult height in a large population

Solution: Normal — a continuous measurement that tends to cluster symmetrically around an average, the classic bell-curve shape.

2.2 For scenario (b), if $p = 0.65$, what is $E[X]$?

Solution: Using $E[X] = np$ for a Binomial:

$$E[X] = 200 \times 0.65 = 130$$

Exercise 3: Regression

Suppose a researcher wants to examine the relationship between parents' years of schooling (X) and their adult child's occupational prestige score (Y) across all U.S. families.

3.1 Write the regression equation using index notation. What does the subscript i represent?

Solution:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

The subscript i identifies a particular adult child or family.

3.2 Interpret the slope coefficient.

Solution: β_1 represents the expected difference in the child's occupational prestige score associated with one additional year of parental schooling.

3.3 Identify the estimand, a plausible estimator, and an example of an estimate.

Solution:

- **Estimand:** the population slope β_1
- **Estimator:** ordinary least squares (OLS)
- **Estimate:** the value obtained from the sample, such as $\hat{\beta}_1 = 1.5$

3.4 Name one plausible factor that could be included in the error term, ε_i.

Solution: Parental income, neighborhood conditions, or the local labor market.